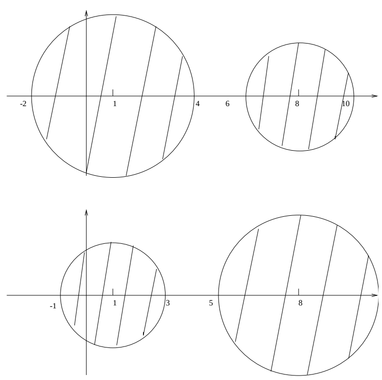
Math 526, S05 **Name:**
Quiz 11 **SSN: xxx-xx-**

<u>Instructions:</u> **There are totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.**

Problem 1. (20 points) Let $\mathbf{A} = \begin{bmatrix} 8 & -2 \\ 3 & 1 \end{bmatrix}$,
a) Use the Gerschgorin Circle Theorem to show that $\mathbf{A}$ has a dominant eigenvalue and give a rough estimate for the location of this dominant eigenvalue.
Solution:

$$a_{11} = 8, \quad a_{22} = 1, \quad r_1 = 2, \quad r_2 = 3, \quad c_1 = 3, \quad c_2 = 2$$



From the above figure, it is easy to see that A has a dominat real eigenvale λ_1 such that $6 \le \lambda_1 \le 10$.

b) With the starting vector $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, compute the first two loops of the power method for the matrix $\mathbf{A}$.
Solution:

1st loop:

$$\mathbf{z} = \frac{\mathbf{x}}{||\mathbf{x}||_2} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{x} = A\mathbf{z} = \begin{bmatrix} 8 & -2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 8 \\ 3 \end{bmatrix}, \quad \alpha = \mathbf{z}^t\mathbf{x} = 8.$$

2nd loop:

$$\mathbf{z} = \frac{\mathbf{x}}{||\mathbf{x}||_2} = \begin{bmatrix} 0.9363 \\ 0.3511 \end{bmatrix}, \quad \mathbf{x} = A\mathbf{z} = \begin{bmatrix} 8 & -2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 0.9363 \\ 0.3511 \end{bmatrix} = \begin{bmatrix} 6.7884 \\ 3.1601 \end{bmatrix},$$

$$\alpha = \mathbf{z}^t\mathbf{x} = 7.4658.$$